We shall now see a few types on constructing these quadrilaterals when its:

(i) 4 sides and a diagonal are given.
(ii) 3 sides and 2 diagonals are given.
(iii) 4 sides and an angle are given.
(iv) 3 sides and 2 angles are given.
(v) 2 sides and 3 angles are given.

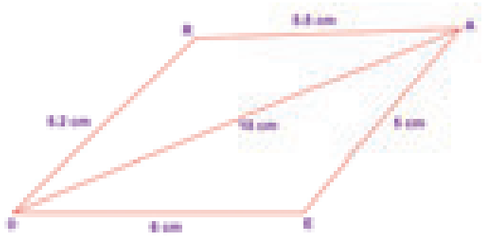
5.10.1 Constructing a quadrilateral when its 4 sides and a diagonal are given

Example 5.21

Construct a quadrilateral *DEAR* with *DE*=6 *cm*, *EA* = 5 *cm*, *AR* = 5.5cm, *RD* = 5.2 *cm* and *DA* = 10 *cm*. Also find its area.

Solution:

Given:
DE = 6 *cm*, *EA* = 5 *cm*, *AR* = 5.5 *cm*,
RD = 5.2 *cm* and a diagonal
DA = 10 *cm*



Rough diagram

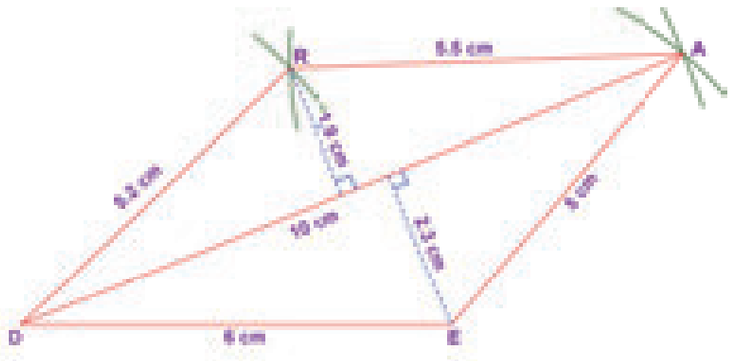


Fig. 5.40

Steps:

4. Draw a line segment *DE* = 6 *cm*.
5. With *D* and *E* as centres, draw arcs of radii 10 *cm* and 5 *cm* respectively and let them cut at *A*.
6. Join *DA* and *EA*.
7. With *D* and *A* as centres, draw arcs of radii 5.2 *cm* and 5.5 *cm* respectively and let them cut at *R*.
8. Join *DR* and *AR* .
9. *DEAR* is the required quadrilateral.

Calculation of Area:

Area of the quadrilateral *DEAR* $= \frac{1}{2} \times d \times (h_1 + h_2)$ *sq. units*

$= \frac{1}{2} \times 10 \times (1.9 + 2.3) = 5 \times 4.2 = 21\ cm^2$

05_8th_Maths_Geometry_Chap 5.indd 182 2/12/2020 12:17:48 PM